PETROPHYSICS | LECTURE 14 | INDEPENDENT READING

Acoustic emission and fracture propagation

A guided tour of experimental evidence in granite, sandstone and carbonate rocks

This guide accompanies the lecture and the UCS laboratory. Read the figures actively: identify the measured quantity, the loading stage and the inference made from the data. You do not need to memorize the papers; you should be able to explain what each measurement reveals—and what it cannot reveal.

Three questions to guide your reading

- WHEN does damage develop? Compare stress–strain histories with AE rate and cumulative activity.
- WHERE does damage develop? Examine relocated AE hypocenters and localization patterns.
- WHAT process is occurring? Compare inferred source mechanisms, ultrasonic velocities and microstructural observations.

Reading map

Rock	Research example	Read for…
Granite	Thompson (2006); Lockner (1991)	Fault nucleation and propagation
Granite / sandstone	King (2021)	Tensile, shear and compactive AE mechanisms
Sandstone	Fortin (2006)	AE and compaction-band localization
Carbonates	Baud (2017); Schubnel (2006)	Detected damage and acoustically quiet damage
Granite	Aben (2019)	AE locations + evolving P-wave tomography

How to use the guide

For each paper: spend 3–5 minutes on the experimental setup, examine the recommended figures, then write a two-sentence answer to the question. Save the final synthesis questions for after your AE-rate analysis.

CASE STUDY 1 | GRANITE

When does a fault nucleate and begin to propagate?

Thompson et al. (2006)

Fracture in Westerly Granite under AE Feedback and Constant Strain Rate Loading. Pure and Applied Geophysics, 163, 995–1015.

Why read it? Track the transition from scattered AE sources to a propagating fracture and compare slow and faster loading.

Experiment. Triaxial compression of Westerly granite at 50 MPa confinement, including tests with AE-feedback loading.

Look at the figures. Figure 4 links differential stress, velocity and AE locations; Figure 6 shows the slow-loaded failure sequence; Figures 10–11 quantify fracture growth and acceleration.

Main finding. AE source positions expose fracture geometry that AE rate alone cannot provide. Near rapid rupture, overlapping signals may prevent reliable event locations.

THINK ABOUT IT Could a rapidly rising AE count distinguish distributed cracking from one growing fault? Explain your answer.

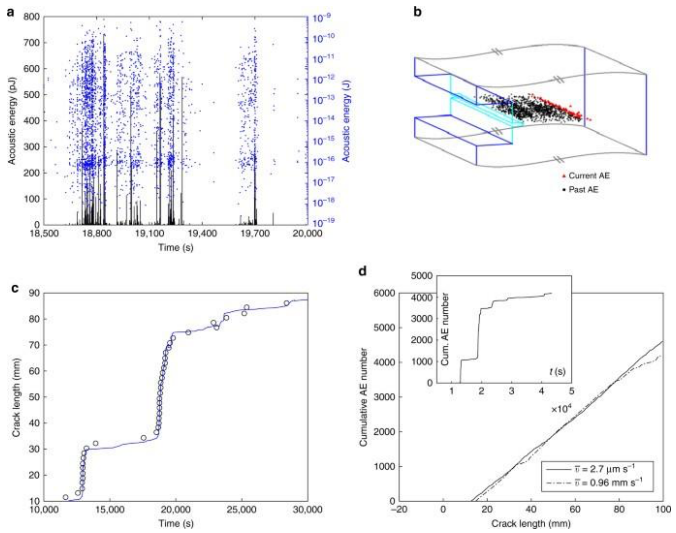


Figure from the original student guide: Thompson et al. (2006). See the source publication for full axes and caption.

Lockner et al. (1991)

Quasi-static fault growth and shear fracture energy in granite. Nature, 350, 39–42. https://doi.org/10.1038/350039a0

Why read it? Understand why AE can be used to control an experiment as well as monitor it.

Experiment. AE-feedback loading slows otherwise unstable fracture propagation in granite.

Look at the figures. Figures 1–2: mechanical loading and the evolving spatial distribution of located AE sources.

Main finding. Fracture develops through a localized, propagating process zone rather than instantaneous failure throughout the specimen.

THINK ABOUT IT Why does controlling AE activity help reveal stages normally hidden by rapid rupture?

What kind of microfracture produces an AE?

King et al. (2021)

Source Mechanisms of Laboratory Earthquakes During Fault Nucleation and Formation. JGR: Solid Earth. https://doi.org/10.1029/2020JB021059

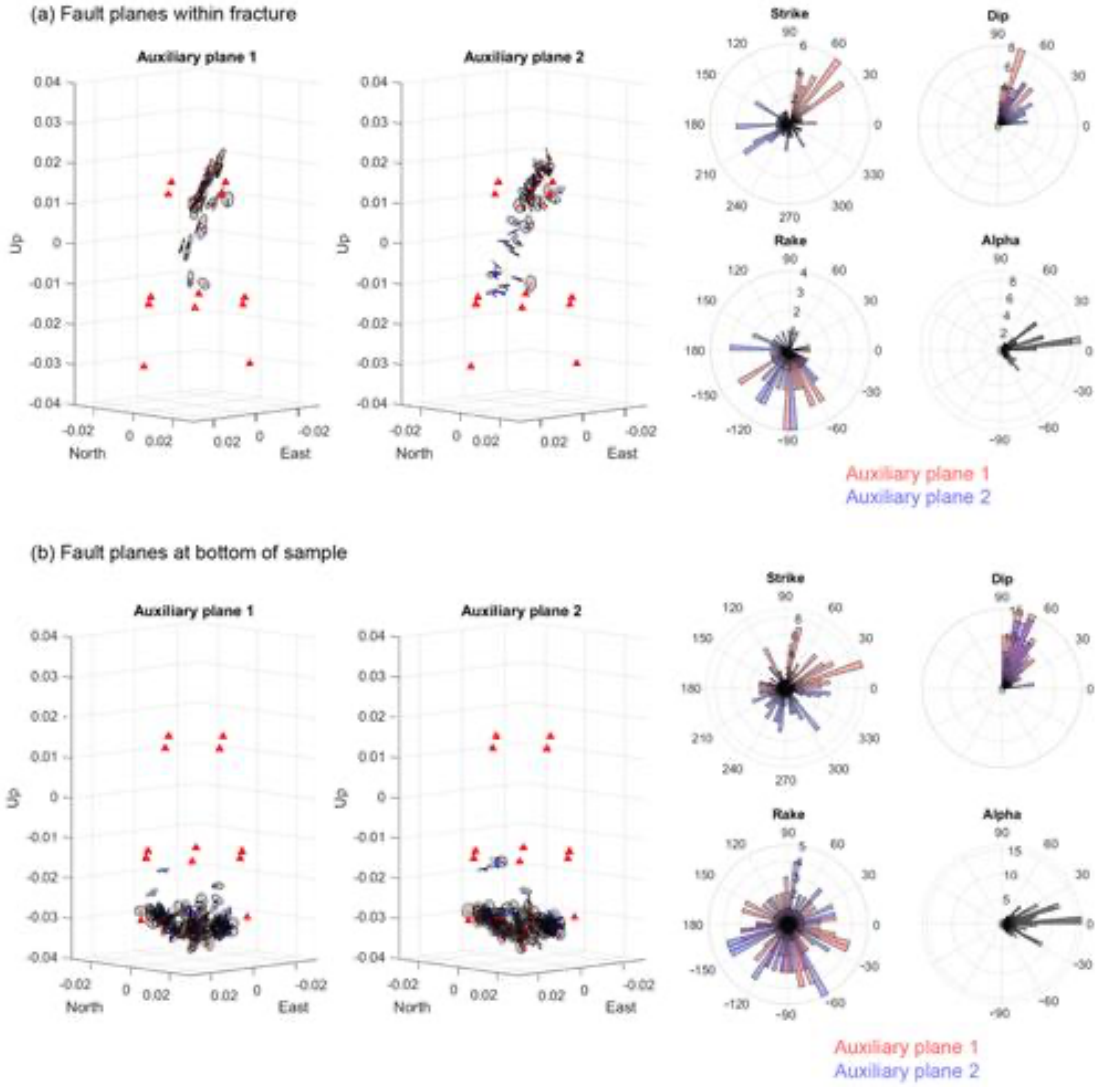
Why read it? Compare opening (T-type), shear (S-type) and compactive (C-type) sources, rather than treating all events as identical tensile cracks.

Experiment. Triaxial tests on Alzo granite and Darley Dale sandstone under 5–40 MPa confinement, using AE first-motion polarities.

Look at the figures. Figures 5–6 display AE source mechanisms and their locations in granite and sandstone; Figure 9 summarizes their evolution.

Main finding. Both rocks generate multiple AE source types. Mechanism proportions and spatial arrangements are affected by lithology and loading.

THINK ABOUT IT Which aspects of an AE record allow you to infer the mechanism of a microfailure, rather than just its occurrence?



Published figure reproduced in your original guide: King et al. (2021). T, S and C interpretations depend on polarity analysis.

CASE STUDY 3 | POROUS SANDSTONE

Can AE reveal compaction bands instead of shear faults?

Fortin et al. (2006)

Acoustic emission and velocities associated with the formation of compaction bands in sandstone. JGR: Solid Earth, 111, B10203. https://doi.org/10.1029/2005JB003854

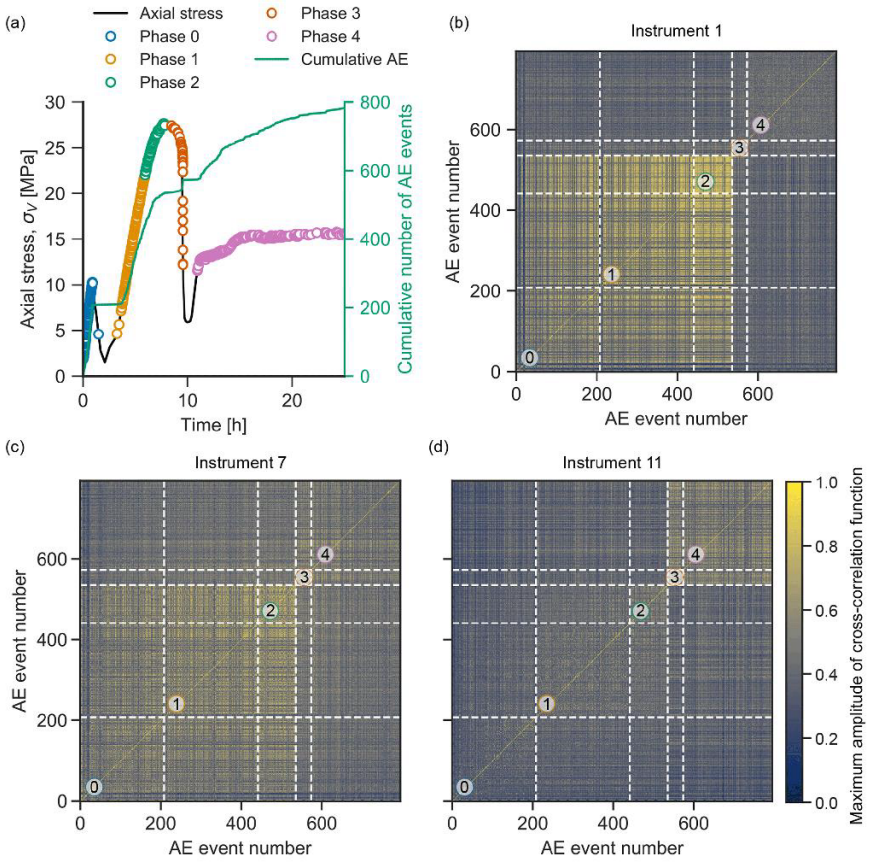
Why read it? Understand localization under pore collapse, where a single inclined shear fracture is not the only possible outcome.

Experiment. Saturated Bleurswiller sandstone compressed triaxially at 60–100 MPa confinement and 10 MPa pore pressure.

Look at the figures. Figure 6 compares stress and cumulative AE; Figures 7–8 show sequential AE locations through the development of compaction bands.

Main finding. AE hypocenters gather within subhorizontal localization structures that are consistent with post-test microstructural evidence.

THINK ABOUT IT What would distinguish an evolving compaction band from a propagating inclined shear fault in an AE-location plot?



Compaction-band observations from the original guide (Fortin et al., 2006).

Related reading — Townend et al. (2008)

Imaging compaction band propagation in Diemelstadt sandstone using acoustic emission locations. Focus on how event location versus time can constrain the motion of a compaction front.

CASE STUDY 4 | CARBONATE ROCKS

Does detectable AE always follow deformation?

Baud et al. (2017)

Inelastic Compaction in High-Porosity Limestone Monitored Using Acoustic Emissions. JGR: Solid Earth. https://doi.org/10.1002/2017JB014627

Why read it? Explore acoustic evidence for compaction and localization in a porous carbonate.

Experiment. Triaxial tests on Saint-Maximin limestone combining strain, AE activity, locations, and ultrasonic measurements.

Look at the figures. Figures 8 and 10–11 compare mechanical response, velocity changes and relocated AE activity under different conditions.

Main finding. Compaction and localizing deformation can produce structured AE distributions, not just a single shear-fault signal.

THINK ABOUT IT How would you combine velocity evolution and AE locations to test an interpretation of localized compaction?

Schubnel et al. (2006)

Transient creep, aseismic damage and slow failure in Carrara marble deformed across the brittle–ductile transition. GRL, 33, L17301.

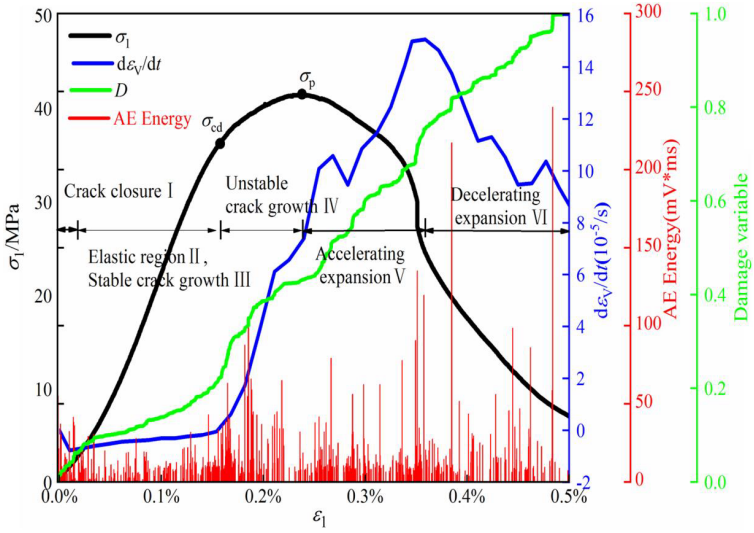
Why read it? Recognize a crucial limitation of AE monitoring: an acoustically quiet specimen may still be deforming and accumulating damage.

Experiment. Triaxial deformation of Carrara marble near the brittle–ductile transition.

Look at the figures. Figure 2: compare the stress–strain record and P-wave velocity evolution during slow deformation.

Main finding. Large velocity changes and slow failure may accompany very little detectable AE radiation in the monitored frequency band.

THINK ABOUT IT Would a low AE rate alone justify concluding that little or no damage occurred? Why not?



Example from the original guide: time-dependent mechanical and acoustic changes in marble. Check the paper for measurement definitions.

CASE STUDY 5 | GRANITE | ADDED READING

How do AE locations and P-wave tomography complement one another?

Aben et al. (2019)

Rupture Energetics in Crustal Rock From Laboratory-Scale Seismic Tomography. Geophysical Research Letters.
https://doi.org/10.1029/2019GL083040

Why read it? Connect the two techniques already introduced in the course: passive AE source locations and active measurements of P-wave velocity.

Experiment. Triaxial deformation of dry Lanhélin granite at 100 MPa confinement. Sixteen piezoelectric transducers monitored AE and active velocity surveys while fracture propagation was slowed using AE-informed load reductions.

Look at the figures. Figure 1: experimental sensors. Figure 2: sequence of Vp tomographic cross-sections with superimposed AE locations during localization, rupture growth and frictional sliding. Figure 3: fault-perpendicular velocity profiles and uncertainties.

Main finding. AE locations delineate the migrating rupture front, while the accompanying low-Vp zone maps damaged, weakened rock around the growing fault. Their spatial patterns are complementary, not interchangeable.

> THINK ABOUT IT In Figure 2, does every low-Vp region contain a corresponding cluster of located AE? What does this tell us about the difference between accumulated damage and radiating events?

What to look for in Figure 2

- Locate the spatial cluster of AE sources at successive loading stages.
- Identify where Vp decreases relative to the intact initial material.
- Compare the arrival of the rupture tip with the formation of the low-velocity zone.
- Explain why AE location maps and tomography cannot be treated as direct images of the same physical quantity.

Helpful quantitative result: within the localized damage zone, Vp decreases by about 25% relative to nearby regions outside the fault; this corresponds to a substantial reduction in P-wave modulus. The authors use these elastic changes to investigate off-fault energy dissipation.

Open-access article: https://doi.org/10.1029/2019GL083040

What can—and cannot—we learn from AE?

Measurement	What it tells you	What remains uncertain
AE rate	Timing and changes in detected activity	Location, source process, detection completeness
AE locations	Spatial migration and localization	Non-radiating damage and location uncertainty
AE polarities / source analysis	Inferred fracture mechanisms	Model assumptions, sensor coverage
P-wave velocity / tomography	Elastic-property changes and damage zones	Unique crack geometry and mechanism

Final reading questions

- The AE count accelerates near peak load. Which interpretations are supported immediately? Which require AE locations or other measurements?
- Granite develops an inclined fracture while sandstone develops compaction bands. Which observable differences would you expect in mechanical, AE and ultrasonic measurements?
- A sample becomes substantially slower for P waves but generates almost no AE. Give two physically plausible explanations.
- Design the minimum additional measurements needed to demonstrate that a fault propagates during a UCS experiment.

Bridge to your laboratory

In the upcoming lab you will compute an AE rate from an event catalogue. Specify whether you count channel hits or associated multi-sensor events, state the time-bin width, and compare the same time base with the mechanical data. Do not interpret the event rate as a direct crack-growth-rate measurement.

SELECTED REFERENCES / LINKS

Thompson et al. (2006), Pure and Applied Geophysics, 163, 995–1015.

Lockner et al. (1991), Nature, 350, 39–42. doi:10.1038/350039a0

King et al. (2021), JGR: Solid Earth. doi:10.1029/2020JB021059

Fortin et al. (2006), JGR: Solid Earth, 111, B10203. doi:10.1029/2005JB003854

Baud et al. (2017), JGR: Solid Earth. doi:10.1002/2017JB014627

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Aben et al. (2019), GRL. doi:10.1029/2019GL083040